# Jennifer Grannen

jenngrannen@berkeley.edu | (720) 244-5121 | github.com/jenngrannen | jenngrannen.com

## EDUCATION

**University of California, Berkeley** — **Berkeley, CA**
*B.A. in Computer Science; Minor in Data Science* — Expected May 2021

- **Overall GPA:** 3.89 / 4.0, **Major GPA:** 3.89 / 4.0
- **Coursework:** Probability Theory, Algorithms, Data Science, Convex Optimization, Data Structures, Discrete Mathematics, Linear Algebra, Computer Architecture, Differential Equations, Artificial Intelligence, Machine Learning, Deep Reinforcement Learning

## RESEARCH EXPERIENCE

**Berkeley AI Research - AUTOLAB** — **Berkeley, CA**
*Undergraduate Researcher advised by Prof. Ken Goldberg* — Aug 2019 – Present

- Developing interpretable representations for perception-driven robotic manipulation of deformable objects.

**UC Berkeley RISELab** — **Berkeley, CA**
*Undergraduate Researcher advised by Prof. Joey Gonzalez & Prof. Ion Stoica* — Sep 2019 – Aug 2020

- Improved calibration of transformer-based methods with data augmentation techniques for NLP applications.

## WORK & TEACHING EXPERIENCE

**UC Berkeley EECS Department** — **Berkeley, CA**
*Teaching Assistant* — Aug 2018 – Present

- TA, CS 188 (Artificial Intelligence) - Fa20, Sp21: Leading office hours and discussion sections, teaching concepts such as Bayes nets, reinforcement learning, and Markov models.
- Lab Assistant, Data 100 (Data Science) - Sp20: Assist in lab sections and office hours.
- Lab Assistant, CS 61B (Data Structures) - Fa18: Assist students in lab sections with assignments and projects.

**Goldman Sachs** — **New York City, NY**
*Securities Strategy Summer Analyst* — May 2019 – Aug 2019

- Franchise Analytics Strats Team: Implemented functionality to recognize new products on existing NLP framework. Greatly improved platform experience, resulting in 10x user increase among traders.
- Equities Flow Volatility Team: Added support for running and displaying risk measures for exotic products on front-facing application. Decreased quote latency by ~33%.

## PUBLICATIONS & PROJECTS

**Untangling Dense Knots by Learning Task-Relevant Keypoints.**
J. Grannen*, P. Sundaresan*, B. Thananjeyan, J. Ichnowski, A. Balakrishna, M. Hwang, V. Viswanath, M. Laskey, J. E. Gonzalez, K. Goldberg. CoRL 2020. *Plenary Presentation (One of 20)*. (https://arxiv.org/pdf/2011.04999)

**Learning Rope Manipulation Policies Using Dense Object Descriptors Trained on Synthetic Depth Data**.
P. Sundaresan, J. Grannen, B. Thananjeyan, A. Balakrishna, M. Laskey, J. E. Gonzalez, K. Goldberg. ICRA 2020. (https://arxiv.org/abs/2003.01835)

**Learning Dense Visual Correspondences in Simulation to Smooth and Fold Real Fabrics.**
A. Ganapathi, P. Sundaresan, B. Thananjeyan, A. Balakrishna, D. Seita, J. Grannen, M. Hwang, R. Hoque, J. E. Gonzalez, N. Jamali, K. Yamane, S. Iba, K. Goldberg. Preprint. (https://arxiv.org/abs/2003.12698)

## ACTIVITIES & OUTREACH

**AUTOLAB Outreach** — **Berkeley, CA**
*Workshop Leader (AI4ALL, SWE, TBP)* — Mar 2020 – Present

**Society of Women Engineers, UC Berkeley** — **Berkeley, CA**
*Instructor, Outreach Lead (SWE++)* — Jan 2020 – Present

**Machine Learning at Berkeley** — **Berkeley, CA**
*Officer (Partner Relations), Blog Writer* — Jan 2019 – Jul 2020

**Expanding Your Horizons, UC Berkeley** (Girls in STEM Conference) **Berkeley, CA**
*Undergraduate Mentor* Aug 2017 – Present

## HONORS, SOCIETIES, & PROFESSIONAL ACTIVITIES

*Paper Reviewer, IEEE International Conference on Robotics and Automation (ICRA) 2021* Dec 2020
*UC Berkeley EECS Honors Program* Jan 2020 – Present
*Upsilon Pi Epsilon (CS Honor Society)* Aug 2019 – Present
*Tau Beta Pi (Engineering Honor Society)* Aug 2018 – Present
*UC Berkeley Dean's Honor List* May 2018

**Juni Learning** **Berkeley, CA**
*Computer Science Instructor* Dec 2018 – May 2019
- Taught introductory programming concepts in Python, Java, and Scratch to grade-school students.